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The digital divide: A review and future research agenda

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ABSTRACT

This article provides a systematic review of the digital divide, a phenomenon which refers to disparities in Information and Communications Technology access, usage, and outcomes. It uniquely identifies the factors affecting the digital divide that have emerged in recent years (2017–2021) as well as investigate if there are new forms or levels of the divide that have surfaced in recent literature. The findings, based on 50 included studies, show that the factors affecting the digital divide can be classified into three different segments and nine main categories: *sociodemographic, socioeconomic, personal elements, social support, type of technology, digital training, rights, infrastructure,* and *large-scale events.* Out of all factors, education has been linked to the digital divide the most. The majority of recent literature have studied Level 2 of the divide. Also, only one article in the sample has considered the digital divide at the firm level. Findings also show that a new form, *type-of-internet access*, and two potential new levels of the digital divide, *algorithmic awareness* and *data inequalities*, have been identified in the contemporary literature. The results contribute to the understanding and development of the different perspectives of the digital divide concept. They also contribute to the stream of literature on the determinants of the divide and to the social inequalities and digital inclusion literature. This review can be seen as a guide for managers to realize and understand the forms that the divide can take and to delve into their organizational capabilities on the digitalization front and evaluate where further development is needed within their organizations to help diminish the divide.

1. Introduction

With the rise of the digital economy, an important and evolving topic has been the digital divide. It was first granted international exposure and brought to the forefront when it began appearing in several United Nations (UN) reports and has become a critical concern for organizations, policymakers, and scholars across various fields (Ganesh and Barber, 2009; Van Dijk, 2020). Digital divide refers to the gap between people who have adequate access to information communication technology (ICT) and people who have poor or no access to ICT (Soomro et al., 2020). The different forms of digital gaps can intensify inequalities within society as they can either restrict or improve citizens' social and economic capital as well as their abilities to participate in society (Ragnedda, 2017). The digital divide has been described as a critical topic for social justice in the twenty-first century (Rogers, 2016) and some scholars have considered it a source of poverty and used the term "digital poverty" (Manduna, 2016; Setthasuravich and Kato, 2020). It is fair to state that the digital revolution that is in progress does not provide the same opportunities for every individual equally and, therefore, creates social inequalities (Bartikowski et al., 2018). ICT is also regarded as one of the most powerful tools to help realize the Sustainable Development Goals (SDGs) initiative launched by the UN and its effective usage has been discussed as a source of contribution to the SDGs' targets, particularly to Goal 10 "Reduced Inequalities" (United Nations, 2015, 2020). Understanding the digital divide concept and its different perspectives and seeing different forms of how not everyone benefits from the technology equally will help in identifying clear technology needs that could lead to the development of more coherent frameworks and policies to address those needs and to intensified efforts to diminish and bridge digital gaps. Goal 10 has been observed as one of the SDGs that are most influenced by ICT since ICT can play a significant role in contributing to equality by allowing access to important information and enabling potent involvement and participation of citizens in their economy as its effective usage is linked to opportunities in education, training, and employability (Kerras et al., 2020).

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Although advances have been made to address the digital divide, it remains ever-present (Centeio, 2017). Additionally, the outbreak of the Coronavirus has had a huge impact on bringing the topic to the forefront. News media, governments, and organizations have emphasized that the concerns about the digital divide have been particularly dire during the pandemic with so many people forced to work, study, access services, and socialize from home. Amid the pandemic, the UN Secretary-General said that “the digital divide is now a matter of life and death” (UN, 2020). The pandemic has forced people to take an exceptional digital leap in everyday life practices (Iivari et al., 2020). This has developed new routines where the world heavily relies on the Internet and digital devices. However, due to lack of resources and effective digital usage, the pandemic has created an exacerbated situation where people who are not well connected to the Internet are encountering exclusion and suffering from other disadvantages (De et al., 2020).

Having become a more digitally dependent world over the past year due to the pandemic, the present is the ideal time to review recent literature on the digital divide and identify how it has progressed with the growing digital economy. Recent literature has seen a renewed interest in the divide with a strong emphasis on empirical work (e.g., Dutton and Reisdorf, 2019; Reddick et al., 2020). To the best of our knowledge, there have only been two articles in the literature since 2017 that have summarized material on the digital divide (Aissaoui, 2021; Vassilakopolou, 2021). One of them only covered the material in one particular field (Information Systems) and the other only looked at some determinants of the divide. Nevertheless, this article is a review that comes with two aims: (1) to study what factors that affect the digital divide have been identified in recent literature, and (2) to investigate if there are new forms or levels of the digital divide that have been identified in recent literature. Examining recent studies is important as the digital divide concept is ever evolving (Soomro et al., 2020). To achieve this, we consider studies from 2017 and onwards. As far as we are aware, this is the first review on the digital divide that seeks to determine new forms and levels of the divide, thus assisting in framing the concept and contributing to its development. It is also the first review that combines the two stated aims. The review, therefore, also contributes to the social inequalities and digital inclusion literature by identifying influences on the divide and possible emerging forms of the concept. To accomplish our aims, we adopt Transfield et al.'s (2003) systematic review approach and follow the structural content analysis approach of Mayring (2014) to formulate synthesis of knowledge on recent digital divide content ($n = 50$) in the data analysis and synthesis stage. This review comes with the aspiration that more management scholars would join global efforts to understand the digital divide and facilitate the exploration of aspects of the divide that have not yet been realized in policies. It will help organizations, governments, and policymakers develop strategies and policies to tackle the divide better. Additionally, this review can help better position work towards the SDGs targets. Identifying needs in relation to digital abilities and competences will have a positive impact on enhancing levels of sustainable development (Hidalgo et al., 2020).

2. Background literature

2.1. Levels of the digital divide

The ‘digital divide’ term gained popularity in policy-making contexts before it was given attention by academic studies in a wide range of disciplines. Academic studies discussing the digital divide began to emerge during the early 1990s. The digital divide concept at that time was considered to be a binary division between people who had access to computers and the Internet and those who did not have that access (Dewan and Riggins, 2005; Hoffman et al., 2000). This access gap, later termed as the ‘first-level digital divide’, was extensively studied that by the beginning of the 20th century, there were more than 14,000 publications that discussed the digital divide issue (Yu, 2006).

Digital divide began to be understood in a broader perspective in the early 2000s when many scholars argued that the concept ought to be interpreted based on several factors and not solely on the division between the “have and have-not” (Leavitt, 2002). Subsequently, the digital divide was construed based on factors such as the accessibility of relevant content, the quality of the Internet connection, and the knowledge and skills of Internet users (e.g., Dewan and Riggins, 2005; DiMaggio et al., 2004; Hargittai, 2002; Van Dijk and Hacker, 2003). The shift in the concept to a focus on differences in digital skills and digital usage was termed the ‘second-level digital divide’ (Dewan and Riggins, 2005; Hargittai, 2002; Van Dijk, 2005). The discussion on users' unequal skills and knowledge of handling Internet devices was also where the term ‘digital inequality’ emerged from (Hargittai, 2002). The second-level digital divide has received a lot of attention from scholars (e.g., Mossberger et al., 2003; Van Deursen and Van Dijk, 2011; Van Deursen et al., 2016), and has been summarized to particularly center around inequalities in technical means, autonomy of use, patterns of use, and skills (DiMaggio et al., 2004). Van Dijk and Hacker (2003) and Van Dijk (2006) also mention mental access within this level of the digital divide as it implies the inequality of motivational access that prevents individuals from using specific technologies. This may comprise of concepts associated with low self-efficacy, computer anxiety, and other psychological factors.

Furthermore, the development of the digital divide concept deviated from the view that access to the Internet and the skills and use of this technology were the only levels of this divide. Several scholars contended that, to approach digital divides more comprehensively, the consequences and outcomes of the Internet use should be considered (e.g., Selwyn, 2004; Van Dijk, 2005). It cannot be assumed that access to and the use of technology automatically provide all the benefits of the technology. The inequality of users' capacities to exploit Internet and communications technology can easily affect individual outcomes and benefits (Ragnedda, 2017). The shift from skills and use of the Internet to a focus on the beneficial outcomes of using the Internet has been labelled the ‘third-level digital divide’ (Wei et al., 2011). This third level of the divide occurs when digital skills and use of the Internet do not lead to beneficial consequences for all individuals (Van Deursen et al., 2016). Thus, the digital divide concept has experienced a significant transformation and has procured a fuzzy definition that now covers ICT access, usage, and outcomes (Shakina et al., 2021). It is apparent that it is progressively considered a dynamic and multidimensional phenomenon caused by an array of factors (Bruno et al., 2011).

The rapid penetration of the mobile Internet in the 2000s led some scholars to argue that the digital divide was shrinking (Stump et al., 2008). However, it is important to highlight that the divide still exists in its three different levels and takes on different forms that determine the equality within the digital world (Kolb et al., 2020). As the world advances and a futures perspective is developed, new ways of thinking of digital divides become relevant. Since the concept has a constantly evolving nature, it is pertinent to know if and how the concept has evolved with the digital revolution over recent years. Knowing this will help organizations and governments adjust and balance their capabilities over time (i.e., be more agile; Shams et al., 2021) as well as set plans to tackle the different forms of the divide that emerge. Hence, one of the questions that this review aims to answer is “*Are there new forms or levels of the digital divide that have been identified in recent literature?*” Such a question is also important to surpass the thinking that there are only three levels of digital divide that take fixed forms, thus opening up opportunities to expand the body of knowledge. We planned this question by considering how the literature is focused on only three levels of the divide and desiring to explore possibilities to expand the subject, especially taking into consideration recent developments in the world (Tranfield et al., 2003). The harsh inequalities in societies have been intensely highlighted as the world struggled to deal with the COVID-19 pandemic (Bapuji et al., 2020), including the digital divide. Asking such a research question is also important as several scholars and policymakers already consider the first level of the divide to have been overcome (Stump et al., 2008), and this embodies an opportunity to research recent literature on the different levels equally, without a bias of researchers towards the latest levels of the divide.

2.2. Determinants of the digital divide

It is not enough to explore the forms and levels of the digital divide. It is also imperative to study what contributes to these levels. The study of their determinants helps to understand how the phenomenon takes place better and will assist those attempting to tackle the divide and striving to achieve equal benefits in society that derive from digitalization. For instance, the literature has found that the link between being digitalized and agility is substantial. Agility is significant in today's changing world and being agile demands many skills. Nevertheless, one of the main determinants of agility has been the country or industry level of digitalization (Škare and Soriano, 2021). Digital competencies have been associated with agility at various levels, such as the individual level (Seale et al., 2010) and the workplace level (Breu et al., 2002). Agility is deemed to be a crucial success element at different levels and the development of digital competencies that will exploit opportunities and respond to change in the environment is needed for this agility to come into being (Chatterjee et al., 2021).

Plenty of studies in the literature have been conducted to identify determinants of the digital divide. The determinants that have been recognized reveal that the digital divide research is mainly limited to sociodemographic and socioeconomic determinants, such as income, age, educational level, ethnicity, and urbanization level (Hidalgo et al., 2020). For example, triggers of the first and second-level digital divides are shown to be related to demographic characteristics such as age, level of education, and ethnicity (e.g., Helsper, 2012; Blank and Groselj, 2014). A more specific example could be the findings by Jackson et al. (2001) that reveal that African Americans who have less home computer access frequently lack self-efficacy and confidence in their ability to use Internet and communication devices. Going beyond socioeconomic demographics, additional contributing factors that have been identified in the literature relate to an individual's digital skills, motivation, culture, and personality characteristics (e.g., Venkatesh et al., 2014). We acknowledge that determining drivers of the digital divide is vital, but we are also interested in determining any factor that might impact the divide, even if positively. Thus, our other research question is "*What factors that affect the digital divide have been identified in recent literature?*" Asking such a question allows us to go beyond just drivers of the divide to cover positive influences, too. This is an important way to speed up tackling the divide. Instead of only focusing on resolving issues with respect to drivers of the divides, scholars, organizations, and governments can also focus on strengthening any of the identified factors that insert positive forces on the divides.

3. Methodology

3.1. Procedure and sample

The systematic review method was chosen to critically assess the body of literature to answer our two research questions "*Are there new forms or levels of the digital divide that have been identified in recent literature?*" and "*What are the factors influencing the digital divide that have been identified recently?*" A systematic approach enhances the quality of the review process by following a transparent and duplicable procedure. Systematic literature reviews help to identify, evaluate, and synthesize the existing body of knowledge on a topic and is, therefore, a good method to meet the research goals of this study (Tranfield et al., 2003). The years 2017–2021 have been considered for this review to gain insights into recent developments in the digital divide field and to explore if and how research has been changing, especially in light of COVID-19. Covering only a few years of existing literature for the purposes and goals of a review is common in systematic reviews (e.g., Vergne and Wry, 2014).

We adopted Tranfield et al. (2003) systematic review approach and proceeded in three stages (1) planning the review (2) collecting data and (3) analyzing the literature and synthesizing the research findings.

In the first stage, the authors came together and agreed on the keywords and the search terms given the diverse definitions and perspectives of the digital divide. We also agreed on what databases to use for the data collection. The first step of stage two included a literature search through the databases EBSCO host's Business Source Complete (BSC) and Scopus which together cover a wide range of social sciences journals. Although many systematic reviews use only one database (e.g., Hanelt et al., 2021), the combination of databases is adopted to strengthen the inclusiveness of the sample of articles and is frequently used in reviews (e.g., Scheerder et al., 2017). This is also one way to make sure that we were being robust. The used databases are recommended for management studies by our Librarian and are considered the most complete sources of business studies commonly used in literature reviews (e.g., Mas-Tur et al., 2020). The search with BSC was conducted using the keyword combinations 'digital divide' OR 'digital inequality' OR 'digital inequalities' OR 'digital gap' OR 'digital division' OR 'digital equity' OR 'digital disparities'. This initial search yielded 5972 documents. Results were then refined to limit the search to the past five years (2017–2021). The search was also limited to include only peer-reviewed full-text journal articles that were written in the English language (Podsakoff et al., 2005). This yielded 226. Likewise, the search within the Scopus database was conducted according to the same criteria as above and limited to social sciences and business articles. The search generated 215 articles. Combined, the total number of articles was 441.

Subsequently, to provide a quality threshold, only articles published in a journal with a ranking in the CABS (Chartered Association of Business Schools) list were kept. This approach assisted in obtaining an

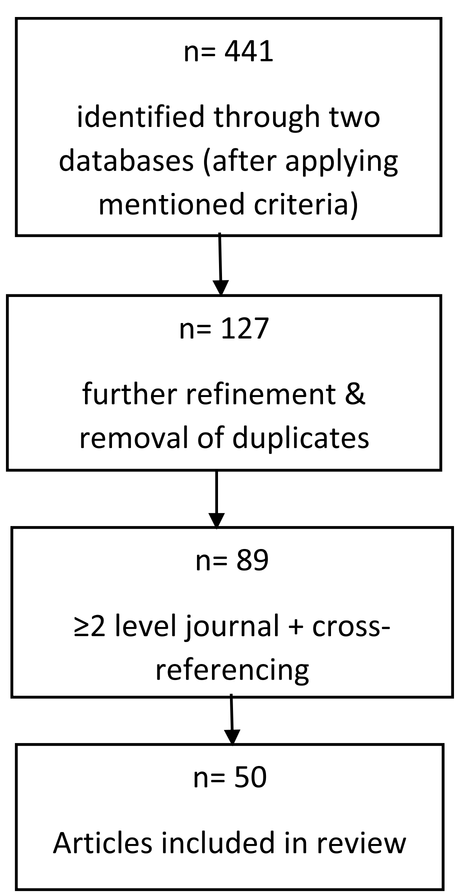

Fig 1. Literature search process.

indication of the journal quality and has been used in systematic reviews (Hiebl, 2021). Nevertheless, for robustness check, we compared our selected list to the Scopus journals to make sure that we did not miss any article in a field that might be underrepresented in the CABS list. Going through some of these articles, it was noticeable that some briefly mentioned our keywords although the paper did not discuss the concepts in any detail. Thus, only articles where the authors considered the digital divide discourse as point of interest were retained (Scheerder et al., 2017). After deleting duplicates across the two databases, 127 articles were left.

We acknowledge that to ensure quality, it is not enough that articles are published in ranked journals and thus we adopted similar ways to other authors and selected only articles published in highly ranked journals (e.g., Bouncken et al., 2015; Kraus et al., 2021). Since the CABS list was used, we maintained articles published in journals with at least a ranking of 2. Thus, another 42 articles were eliminated and 85 were maintained. Citations in the articles published in the top journals were cross-referenced for titles to make sure that we had all relevant articles to be assessed. The retained manuscripts, combined with those identified via cross-referencing, resulted in 89 papers. The final step included reading all the remaining articles and excluding the ones that do not meet the purpose of this review, which condensed the sample into 50 academic journal articles (process demonstrated in Fig. 1).

Data-extraction forms that include details of the articles considered (title, authors' names, year of publication, journal, context of the study, type of data, and applied method) were created. Tranfield et al. (2003) state that this step is important in reducing human error and bias and serves as the data-repository from which the analysis will arise. In our data-extraction forms, we also included the aim, the summary of the main findings, and the level of analysis of each article. Additionally, we entered which level of the digital divide the article referred to.

3.2. Data analysis and synthesis

Stage 3 of the review process included data analysis and synthesis (Tranfield et al., 2003). We followed the structural content analysis approach of Mayring (2014). This is an established analysis approach (e.g., Fastenrath and Braun, 2018; Hanelt et al., 2021) and allows the systematic reduction of large amount of text data from any document by classifying the information into unifying categories (Hanelt et al., 2021). Mayring (2014) recommends undertaking five main steps: (1) constructing a category system based on the research questions, (2) coding relevant parts in the text in line with the category system, (3) revising the created classification framework, (4) coding the text according to the revised category system, and (5) interpreting and discussing the findings.

Based on our research questions, we adopted 'Factors affecting the digital divide' and 'Forms of digital divide' as the initial categories. Subsequently, we extracted the constructs and key findings from each article and allotted them to the two categories. After placing the material into the two categories, we revised the category system by subdividing the existing categories (Mayring, 2014). For example, the 'Factors affecting the digital divide' category was divided into the sub-categories of *sociodemographic influences, socioeconomic influences, personal elements, social support, type of technology, digital training, rights, infrastructure,* and *large-scale events.* After applying the sub-categories, we proceeded to revise the category system by splitting up the sub-categories into the relevant variables of our sample. For instance, the *personal elements* sub-category includes variables such as *trust, motivation, privacy concerns, risk perceptions, values*, and *beliefs*. This is illustrated in Fig. 2 in the following section where we report the findings of the review.

4. Findings

4.1. Sample demographics

Publications each year: 17 out of the 50 considered articles were published in the year 2020, followed by 12 articles published in 2017.

Participants: The included research involved participants from 24 different countries (context is included in the table in Appendix A).

Research methods: The distribution of the considered studies indicates a clear dominance of quantitative research (38 studies). 23 of these studies used a survey and questionnaire approach. 9 studies are qualitative, and 3 papers used a mixed methods approach. 5 of the quantitative articles utilized moderators or mediators in their models.

Data: 24 studies used secondary data for their analysis. The distribution also shows that 9 studies adopted a longitudinal data approach.

(A list of the sample is found in Appendix A)

We synthesize the findings of our review into the framework illustrated in Figs. 2 and 3 which organizes our main results and links together the different insights into a coherent whole (Hanelt et al., 2021). The figures comprise the two aforementioned categories 'Factors affecting the digital divide' and 'Forms of digital divide'. We explain the categories, their corresponding sub-categories, and the key topics of each sub-category in the sections below.

4.2. Factors affecting the digital divide

The category 'Factors affecting the digital divide' represents any antecedents and drivers of the digital divide. It includes any factor that creates or exacerbates the digital divide as well as any factor that contributes to the diminishing of the divide. Our review found that *sociodemographic, socioeconomic*, and *personal elements* as well as *social support, type of technology, digital training, rights, infrastructure,* and *large-scale events* affect the digital divide. We have identified the sociodemographic elements as age, race/ethnicity, gender, population density, geographic disparity, urbanization, urban/rural dimension, remoteness, and size of the country (Scheerder et al., 2017). Among these variables, we found that age has recently been mostly linked to the digital divide (14 times), particularly to Levels 1 and 2 of the divide, followed by gender (5) which has been linked to Level 1 mainly. However, out of all the variables in this category, education, which is part of the *socioeconomic* sub-category, has mostly been associated with the digital divide (15 times). For example, van Ingen and Matzat (2018) found educational differences in the mobilization of online problem-focused coping resources (Levels 2 and 3). Other variables in this sub-category include employment status, occupation, and income (Scheerder et al., 2017).

The third sub-category, *personal elements*, consists of trust, motivations, privacy concerns, risk perceptions, values, attitudes and beliefs, and religion. There is not a single variable that dominates in this classification. Trust was linked twice to the second level of the digital divide, and privacy concerns were drivers of the first and second levels of the divide. Goncalves et al. (2018) found that four basic values (achievement, hedonism, benevolence, and universalism) had a significant impact on the acceptance and adoption of ICT and, thus, offered new insights into the digital divide subject. The fourth sub-category that our review identified is *social support* which consists of access to support, social interaction, and social connections. Welser et al. (2019) found that access to social support is associated with digital skills enhancement (Level 2 of the digital divide). Nevertheless, Helsper et al. (2017) found that there are no real inequalities in access and use of support, but that the inequalities lie in the quality of the support that an individual accesses.

The fifth sub-category of 'Factors affecting the digital divide' that this review identified is *type of technology* that include the computer/mobile dimension, overreliance on smartphones, and lack of equipment. Correa et al. (2020) revealed that people who only access the Internet through mobiles tend to have lower levels of skills and conduct less

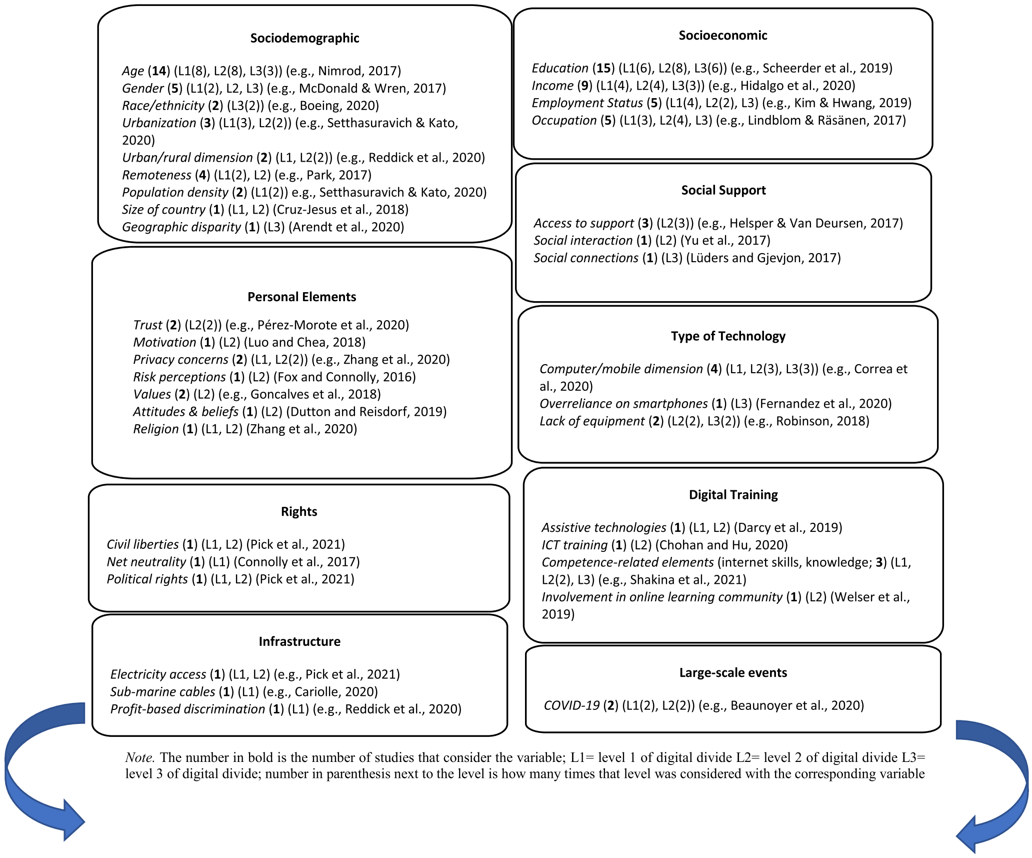

Fig 2.

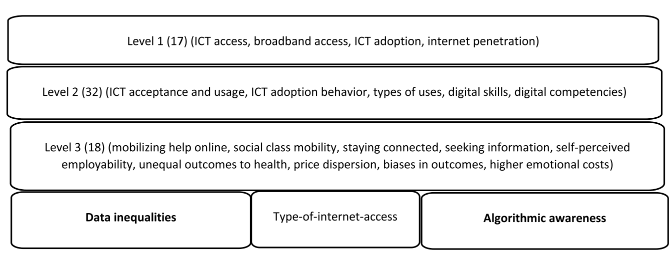

Fig 3. Digital Divide.

diverse activities of the web compared to people who also use the computer. Another example is Fernandez et al. (2020)'s study which found that using mobile internet is not as beneficial as using big devices that allow people to read, write, and create more complex content as mobile internet limits access to content that was not designed to be mobile-friendly, such as access to many databases. The authors also found that people who depend highly on mobiles are less likely to use the Internet in ways that can combat socioeconomic divides (Level 3 of the divide).

The sixth sub-category of the factors that affect the digital divide that we identified is *digital training*. The key variables that fall into this classification are assistive technologies, ICT training, competence-related elements (such as knowledge of technology and internet skills), and involvement in online learning communities. This finding shows that variables associated with Level 2 of the digital divide (skills) can subsequently shape or affect the third level of the divide and, thus, we have included them in the 'factors that impact the digital divide' category. For example, Chohan and Hu (2020) found that ICT training programs increased self-efficacy regarding e-government applications. Moreover, the seventh sub-category that was identified in the review is *rights*. This classification includes civil liberties, political rights, and net neutrality that have been associated with the first and second levels of the digital divide. For instance, Pick et al. (2021) showed that civil liberties and political rights affected the adoption and use of ICT in Latin America and the Caribbean countries.

Another sub-category of the factors affecting the digital divide is *infrastructure*. The classification contains infrastructure-related topics such as electricity access and sub-marine cables (Shannon and Smets, 2010). It also includes profit-based discrimination where broadband carriers choose not to enter a specific area because they do not believe that it will be profitable. For example, Reddick et al. (2020) showed that profit-based discrimination had a negative impact on broadband access. As expected, the infrastructure variables mainly impact the first level of the divide. Finally, the last sub-category that we identified from the most recent literature in our sample relates to *large-scale events*. Beaunoyer et al. (2020) and Iivari et al. (2020) explain how COVID-19 has exacerbated and reinforced a variety of digital divides related to all three levels of the divide. Beaunoyer et al. (2020) describe how, because of the pandemic, issues like unsuitable equipment, having to share equipment, internet traffic, and difficulty in access are aggravated.

The review finds that all the variables in the *personal elements* and *social support* categories have been studied at the individual level of analysis, and that all variables in the *rights* category have been measured from the country level of analysis. Nevertheless, the other categories had a mix between the individual, regional, and country levels of analysis. Out of all the variables, only one, competence-related elements, in the *digital training* sub-category, considered the firm level of analysis.

The remaining results of the factors affecting the digital divide are illustrated in Fig. 2.

4.3. Forms of digital divide

The category 'forms of digital divide' represents the different forms that the digital divide has taken which have helped shape the concept into three levels (Scheerder et al., 2017). These three levels make up the sub-categories for the forms of the divide. Our review found that Level 1 of the divide was adopted in 17 studies, Level 2 was adopted in 32 studies, and Level 3 was considered in 18 studies. Some studies used a combination of two or of all three levels when considering the digital divide (e.g., Huang et al., 2017; Pick et al., 2021; Zhang et al., 2020). Examples of variables in the studies that belong to Level 1 of the divide include ICT access (e.g., Setthasuravich and Kato, 2020) and internet penetration (e.g., Cariolle, 2020). Level 2 includes variables such as digital skills (e.g., Welser et al., 2019) and ICT usage (e.g., Luo and Chea, 2018). Additionally, examples of variables in the considered studies that belong to Level 3 of the divide involve mobilizing help online (van Ingen and Matzat, 2018) and higher emotional costs (Huang et al., 2017). This is illustrated in Fig. 3. Deciding which variables to include in each sub-category was based on the rich existing literature on the levels of the digital divide (Scheerder et al., 2019; Van Deursen et al., 2016). After doing this, the review identified three variables that do not currently properly exist in any of the known levels or do not fit into any of them.

The bottom row of Fig. 3 shows the new forms and level of the digital divide that were identified. Linked to the sub-category *type of technology* from Fig. 2, Bartikowski et al. (2018), who affirm that the nature and scope of the digital divide continues to evolve over time, believe that another important element of the divide is the 'type-of-internet-access'. For example, although smartphones increase digital inclusion, they cannot entirely substitute for the usability and comfort of large screen devices with keyboards and higher processing power. Therefore, the authors compare mobile use to regular internet use, and expand previous research on digital divide by concentrating on the type-of-internet-access. They find that mobile, compared to regular-only, internet access is positively associated with perceived economic situation which, in turn, positively affects life satisfaction. They conduct their analysis on an individual level as well as on a country level and find that the positive effect of mobile internet access, as compared to regular-only, on perceived personal economic situation and life satisfaction is stronger in poorer nations than in richer nations.

Additionally, according to Cinnamon (2020), "conventional understandings of digital inequalities are not always sufficient to explain and address some causes, forms, and consequences of emerging inequalities of the data revolution" (p. 215). The data revolution is generating various forms of inequality that are not captured by core digital divide concepts. Thus, 'data inequalities' exist and are part of digital inequalities but have not been addressed yet.

Gran et al. (2021) question a new form of digital divide. Considering the effects of algorithmic systems, both beneficial and harmful, on people's everyday lives and information access, knowledge of algorithms has the potential to be a crucial issue. If algorithms play a critical role for the participation in public life, then lack of awareness of them appears to create a democratic challenge. Gran et al. (2021) form an algorithmic awareness typology that classify people into six groups: the unaware, the uncertain, the affirmative, the neutral, the skeptic, and the critical. They state that "Being aware of and navigating consciously on the Internet infrastructure could be seen as a new and reinforced level of digital divide" (p. 13). The authors question whether being aware of algorithms or not relates to a new reinforced digital divide.

5. Discussion

The findings answered our research question "What factors that affect the digital divide have been identified in recent literature?" Results show that the factors which influence the divide can be grouped into nine different categories. Nevertheless, the findings also reveal that the factors that determine the digital divide in recent studies can be divided into three different segments: studies that have emphasized established determinants (e.g., educational level and type of occupation), studies that have found variations in an established determinant (e.g., age), and studies that have identified new determinants of the divide. An example of the second segment includes Luders and Gjevjon (2017)'s study that revealed variations within the same elderly group and the digital divide. Older people who were already socially well-connected benefited from online communication more than those who were not. Álvarez-Dardet et al. (2020) also revealed relevant heterogeneity within the older adult group when it comes to ICT adoption. As an example of the third segment, new determinants, Dutton and Reisdorf (2019) found that patterns of attitudes and beliefs shape individuals' behavior online as well as their motivations for getting online.

The findings show that the level of education remains dominant in recent literature as a main factor that influences the digital divide. This factor was deemed to play a very important role in digital divides in past literature (van Deursen and van Dijk, 2014). The results also reinforce the conception that level 2 of the divide factors can influence level 3 of the divide. This is important in order to acknowledge that although the literature has separated the concept into three levels, these levels are strongly linked and should often be considered together for a coherent representation of the digital divide.

An interesting observation that derives from the findings is that *personal elements* mostly impact the second level of the digital divide but have not been tested to be linked to the third level of the concept in recent literature. For example, privacy concerns may affect the outcomes of using ICT and such a finding would therefore make a contribution to the digital divide concept. Similarly, *social support*, which the findings showed is an emerging factor that impacts digital divides, has been associated with levels 2 and 3 of the divide but not with the first level. Thus, there is room to progress research on the digital divide as well as to further contribute to the digital inclusion literature.

The findings also show that although scholars have studied the digital divide at an individual, regional, and national level, only one recent article has considered it at the firm level (Shakina et al., 2021). This is important to realize as studying the concept at the corporate level, for example, can explain gaps in digital resources provision of a firm (Shakina et al., 2021).

Furthermore, the findings answered our second research question "Are there new forms or levels of the digital divide that have been identified in recent literature?" One new form and two potential new levels of the divide emerged in the results. The findings show that type-of-internet-access plays an important role in the digital divide but has not been allocated a confirmed place within the concept yet. Based on the nature of the variable, we see that this form of digital divide fits into

Level 1 (Van Dijk, 2017) as it is not only a matter of access, but also what type of access. Based on Cinnamon (2020) and Gran et al. (2021)'s studies, there is a massive opportunity to explore algorithmic awareness and data inequalities as a fourth level of the divide. This finding is particularly important to expand the digital divide concept. The finding to this research question also shows that Level 2 of the divide has been discussed the most in recent literature.

The findings support the different perspectives of the digital divide and suggest that these perspectives expand beyond solely three levels of the divide. Recognizing that researchers look at the divide from different angles as well as understanding that the concept can take different forms- old and emerging- can help scholars better link the concept to other important elements in the literature. Essentially, understanding the divide in a complete and thorough manner will help in furthering the linkage between the different forms of digitalization and agility. It has already been made clear in the literature that agility is an outcome necessary for survival, resilience, and performance in today's competitive and turbulent world (Miceli et al., 2021; Škare and Soriano, 2021; Troise et al., 2022) and that this agility depends on the investment in digitalization (Lugwisha, 2021; Škare and Soriano, 2021). New capabilities related to digitalization have to be embraced to become increasingly agile (Troise et al., 2022). We mention this association as scholars have recently called for contributions to this important relationship (e.g., Troise et al., 2022).

6. Practical implications

The topic of the digital divide is of great interest to managers as they attempt to cope with escalating uncertainty and volatility in today's market and become more agile (Çallı and Çallı, 2021; Škare and Soriano, 2021). This study can be seen as a guide for managers to delve into their organizational capabilities on the digitalization front. As the relationship between digitalization and agility is substantial, by understanding the divide more, organizations can adjust and balance their capabilities over time to make sure that they remain agile (Shams et al., 2021). Increasing their efforts to develop and improve the digital capabilities and skills will build agility which scholars have argued is the key strategy to counter many challenges in the ever-changing environment and remain flexible in facing new challenges (Shams et al., 2021), and thus constitutes a crucial factor in determining an organization's success (Çallı and Çallı, 2021; Škare and Soriano, 2021). By understanding the factors that affect the digital divide and the different forms that the divide can take, managers can evaluate where further development is needed within their organizations.

This review helps managers to realize that providing good internet access and connectivity only does not necessarily yield positive and equal outcomes for everyone within the company. The findings show managers that older employees may need more support to thrive at work. Also, recognizing the urban-rural differences in access and connectivity, managers will understand that employees living outside the city and who have a long commute may not be able to work from home as efficiently as other employees if asked to work from home or in case another large-scale crisis occurs in the future. Additionally, considering that education affects the digital divide will help managers acknowledge the importance of supporting employees who wish to acquire additional education, a benefit that some organizations already offer.

Since social support and digital training also play a big role in the divide, managers might be encouraged to provide better quality training to their employees as well as support them socially. Organizations should provide support that goes beyond IT services to their employees and offer training that goes beyond enhancing basic digital skills. The provided training should be customized to cater to the needs of the different societal groups.

From the finding that trust, privacy concerns, and risk perceptions influence the digital divide, managers may wish to ease the employees' concerns by assuring them that they respect their privacy (i.e., alleviating privacy concerns) and building a trustworthy relationship which can be done in several ways such as creating an inclusive and fair culture and practicing consistency (Pless and Maak, 2004). Moreover, since attitudes and beliefs have an impact on the divide too, managers can help positively shape and influence the employees' beliefs of using digitalization to achieve their highest potential at work.

Since this review also shows that one of the emerging forms of the digital divide is the type-of-internet-access and that some people tend to adopt an overreliance on accessing the internet through their mobiles (e.g., Fernandez et al., 2020), organizations can provide their employees with larger screens for increased productivity both at work and home, especially during the pandemic. As the pandemic contributed to the digital divide, organizations should always be prepared for the next crisis or further lockdowns. For example, companies should replace any of their outdated equipment that employees use.

There is also a lot that organizations can do to help the community along with the government. Firms can lend a hand to the government in providing suitable equipment for disadvantaged groups. For individuals of the community with no easy access to a good Internet connection, organizations can offer them safe spaces to access the Internet. These spaces can then also be used if the individuals only have access to outdated digital devices. Companies should also engage in financial support, public funding and donations for low-income households. For example, organizations can donate as well as encourage donations of used devices to charities that would distribute them to the digitally excluded people of the community. They can also donate devices such as laptops to schools and universities where students are experiencing a lack of digital devices (Kelley and Sisneros, 2020).

Companies can also assist governments in arranging cheaper home Internet subscriptions within disadvantaged communities to emphasize the significance of what can be accomplished with a fixed connection. This is not to say that a fixed connection is always better than a mobile connection, but it is important to have access to a variety of digital services through different devices as each device is optimized for specific uses which renders device variety important in enhancing digital skills overall (Lamberti et al., 2021). In addition to this, organizations, governments, regulators, internet service providers and content providers should come together to increase broadband service in rural areas.

Another thing to consider is that organizations can help the community through conducting Data Walks discussions, an idea that comes from a pedagogical technique and constitutes an interactive way for local government officials, researchers, program administrators, community stakeholders (including inhabitants), and service providers to take part in conversation revolving around research findings about their community. This can provide an opportunity to co-design activities associated with civic technology and constitutes an important social implication as it is a way to enhance social participation of non-tech savvy citizens. This can also be beneficial for older adults to be co-designers and combat stereotypical ideas about aging. Instead of thinking that the digital divide is always going to be there, i.e., normalizing it, researchers, practitioners, and policymakers should instead talk about it more and problematize it.

There is a need to encourage special policies to be directed toward certain sociodemographic or socioeconomic groups. Referring to Hodge et al. (2017), who found that the digital divide in their study involving older people was not a result of unequal access to technology or socioeconomic disadvantages but the result of the failure of service providers to adapt and accommodate their online engagement strategies to the particular needs of older residents, illustrates the need for policymakers to enhance strategies related to online engagement.

7. Future research directions

By reviewing the literature on the digital divide, it is evident that there is a lot of potential to advance the digital divide concept. First, the new form of the digital divide, type-of-access, should be further explored. One way that this can be done is by studying how this type of access gap differs among countries rather than among societal groups (Bartikowski et al., 2018). It would also be beneficial to study whether this kind of digital divide exists within the same organization among people with different roles or occupations. According to our review, the digital divide has barely been applied within the organizational context. Second, the suggested new levels of the digital divide, algorithmic awareness and data inequalities, should be explored. This presents a huge opportunity to contribute to a new level of the divide which will be considered an enormous accomplishment for the digital divide literature. These two subjects are also deemed 'hot topics' today.

Third, as our findings show that personal elements as factors that influence the digital divide have not been linked to the third level of the divide yet, future research should explore this area that would contribute to Level 3 of the digital divide as well as to the concept as a whole. For example, researchers can study the impact of privacy concerns on ICT outcomes. Additionally, as social support has not been associated with the first level of the divide in recent literature, there is room to make a contribution in relation to this. This is particularly interesting to study amid a pandemic as it would be inevitably harder to obtain support. Research on digital divide in light of COVID is at its infancy and there is a lot of potential to contribute to it.

Fourth, there is still potential for future research to progress the literature on the determinants of the digital divide. This review has found that there is an emerging interest in the variations of sociodemographic factors rather than just considering them as fixed factors that influence the digital divide homogenously (Álvarez-Dardet et al., 2020; Luders and Gjevjon, 2017). The review has also disclosed a recent interest in attitudes and beliefs as new determinants of the digital divide. Thus, the outlook for future research investigating these emerging elements with the digital divide looks promising.

Fifth, future research can also explore consequences of the digital divide (not ICT outcomes). When we were searching through the articles, we noticed a lack of studies that consider the consequences of the concept. Beaunoyer et al. (2020) explored the interplay between COVID-19 and digital inequalities. They argued that on one hand, the crisis worsened digital inequalities, but on the other hand, digital inequalities depicted a major risk for exposure to the virus. For example, when people were not able to go online and lacked alternatives for essential activities such as grocery shopping during the crisis, these people were more likely to expose themselves to the virus to realize essential errands. Exploring similar important relationships will contribute to the nearly non-existent literature on the divide and its consequences.

Sixth, as this review found that the factors affecting the digital divide were mainly explored at an individual, regional, or national level, and that only one study explored it at a firm level (Shakina et al., 2021), future research can address this gap by conducting digital divide studies at a corporate level. Additionally, researchers interested in conducting future reviews on the divide can specify a level of analysis to be considered. For example, a review can be conducted only on the micro level of analysis or on the team level of analysis. A narrower focus could also entail looking at specific sectors in the review such as healthcare or the higher education. Scholars interested in future reviews on the digital divide can also place more emphasis on geography. It might be worth investigating that there is no over-representation or sub-representation of specific areas. Some areas might be more affected by the digital divide and studying this would shed further new light on the topic.

8. Conclusion

The digital revolution does not offer the same opportunities for every individual equally (Bartikowski et al., 2018). To tackle this, we should constantly study the ways in which the digital divide is appearing and what influences it. Therefore, in this study we set out to provide a recent (2017–2021) review on the digital divide that identifies (1) factors influencing the digital divide at different levels and (2) new forms and levels of the divide. Recent literature lacks endeavors that structurally analyze and critically review the digital divide problem. This is the first review that seeks to determine what influences the divide whilst also seeking to determine new forms and levels of the concept. We found that nine main categories of variables affect the divide. Sociodemographic and socioeconomic variables are dominant, affecting the divide at three different levels. Personal-related elements affect the divide at mainly the second level and has not been linked to the third level. Moreover, by linking our findings to the established knowledge on digital divide, we diagnosed a new form of the divide at the first level (type-of-internet-access) and two variables that go beyond the existing levels and form a fourth level which could make a substantial contribution to the development of the digital divide concept. Through these findings, an agenda for future research arose along with important managerial implications. Understanding the divide and finding ways to enhance digital capabilities and skills will help in building agility which is crucial for an organization's success (Çallı and Çallı, 2021; Seale et al., 2010; Škare and Soriano, 2021). This review would help organizations, governments, and policymakers develop strategies and policies to better confront the divide. It is vital that strategies are assessed, adjusted, and created to develop ways that generate value and ensure that technology creates inclusive, prosperous, and sustainable societies.

Although this review was conducted diligently, it is subject to several limitations. First, the process of coding the articles was performed by hand, which could include subjectivity even though we attempted to prevent this by relying on multiple coders and rounds of scrutinizing the coding (Hanelt et al., 2021). Second, articles from the year 2021 are part of the sample although the year has not come to an end yet and thus these articles are not reflective of the whole year. Third, despite being carefully applied, the methodological approach could be criticized for eliminating certain articles. A fourth limitation is that we only included articles that were published in the English language. Fifth, although the contexts of the studies were mentioned, the review did not investigate whether there were some regions that were over-represented in our chosen sample and a recommendation based on this has been made in the future research directions section. Finally, this review investigated the different levels of analysis considered in the sample to see which level was applied the most and which level needs further attention. Nevertheless, this can also be considered a limitation as a narrower approach which entails specifying a particular level of analysis can imply other details that will also shed light on the topic.

Authors statement

We declare that our manuscript is original and has not been previously published or copyrighted. It is also not under consideration for publication in the identical or substantially identical form in any of other peer-reviewed media. We declare that our manuscript meets all requirements of your esteemed journal and there is no conflict of interest or financial support which could have influenced its outcome.

Appendix A

List of the sample

Authors	Year of publication	Journal	Context	Method	Data
Afshar Ali, Alam & Taylor	2020	*Economics of Innovation and New Technology*	Australia	quantitative	secondary
Álvarez-Dardet, Lara & Perez-Padilla	2020	*Computers in Human Behavior*	Spain	quantitative	secondary
Arendt, Haim & Scherr	2020	*Social Science & Medicine*	Germany, Switzerland, Austria, and Belgium	quantitative	secondary
Barrantes Cáceres & Cozzubo Chaparro	2019	*Information, Communication & Society*	Buenos Aires, Lima, and Guatemala City	quantitative	secondary
Bartikowski, Laroche, Jamal & Yang	2018	*Journal of Business Research*	21 countries	quantitative	secondary
Beaunoyer, Dupéré & Guitton	2020	*Computers in Human Behavior*		theoretical	
Boeing	2020	*Environment and Planning A: Economy and Space*	USA	quantitative	secondary
Bol, Helberger, & Weert	2018	*The Information Society*	Netherlands	quantitative	primary
Burtch & Chan	2019	*MIS Quarterly*	USA	quantitative	secondary
Campos, Arrazola, & de Hevia	2017	Economics of Innovation and New Technology	Spain	quantitative	secondary
Cariolle	2020	*Information Economics and Policy*	45 sub-Saharan African countries	quantitative	secondary
Chohan and Hu	2020	*Information Technology for Development*	China	quantitative	primary
Cinnamon	2020	*Information Technology for Development*		theoretical	
Connolly, Lee & Tan	2017	*Review of Industrial Organization*	USA	qualitative	primary
Correa, Pavez, & Contreras	2020	*Information, Communication & Society*	Chile	quantitative	secondary
Cruz-Jesus, Oliveira, & Bacao	2018	*Journal of Global Information Management*	45 countries	quantitative	secondary
Darcy, Yerbury, & Maxwell	2019	*Information, Communication & Society*	Australia	qualitative	primary
Dutton & Reisdorf	2019	*Information, Communication & Society*	Detroit	quantitative	primary
Eynon, Deetjen & Malmberg	2018	*The Information Society*	Britain	quantitative	secondary
Fernandez, Reisdorf & Dutton	2020	*Information, Communication & Society*	Detroit	mixed	primary
Fox and Connolly	2018	*Information Systems Journal*	United States and Ireland	mixed	primary
Goncalves, Oliveira & Cruz-Jesus	2018	*Computers in Human Behavior*	Angola	quantitative	primary
Gran, Booth & Bucher	2021	*Information, Communication & Society*	Norway	quantitative	primary
Helsper & Van Deursen	2017	*Information, Communication & Society*	Netherlands	quantitative	primary
Hidalgo, Gabaly, Morales-Alonso & Urueña	2020	*Technological Forecasting and Social Change*	Spain	quantitative	secondary
Huang, Cotten & Rikard	2017	*Information, Communication & Society*	USA	quantitative	primary
Humphry	2021	*Information, Communication & Society*	Australia	mixed	primary
Hwang & Nam	2017	*International Journal of Consumer Studies*	South Korea	quantitative	secondary
Iivari, Sharma & Ventä-Olkkonen	2020	*International Journal of Information Management*	Finland and India	qualitative	primary
Kim & Hwang	2019	*Information, Communication & Society*	Korea	quantitative	secondary
Lindblom & Räsänen	2017	*The Information Society*	Finland, the United Kingdom, and Greece	quantitative	secondary
Loh & Chib	2019	*Information Technology for Development*	Singapore	quantitative	primary
Luders & Gjevjon	2017	*The Information Society*	Oslo	qualitative	primary
Luo & Chea	2018	*Information Technology & People*	Rural Cambodia	qualitative	primary
McDonald & Wren	2017	*Oxford bulletin of economics and statistics*	United Kingdom	quantitative	secondary
Nimrod	2017	*Information, Communication & Society*	9 European countries	quantitative	secondary
Okunola, Rowley & Johnson	2017	*Government Information Quarterly*	Nigeria	quantitative	primary
Park	2017	*Journal of Rural Studies*	Rural Australia	quantitative	secondary
Pérez-Morote, Pontones-Rosa & Núñez-Chicharro	2020	*Technological Forecasting and Social Change*	27 European countries	quantitative	secondary
Pick, Sarkar, Parrish	2021	*Information Technology for Development*	Latin America and the Caribbean countries	quantitative	secondary
Reddick, Enriquez, Harris, & Sharma	2020	*Cities*	San Antonio	quantitative	primary
Robinson	2018	*Information, Communication & Society*	California	qualitative	primary
Scheerder, Van Deursen, Van Dijk	2019	*The Information Society*	Netherlands	qualitative	primary
Setthasuravich & Kato	2020	*Transport Policy*	Thailand	quantitative	secondary
Shakina, Parshakov & Alsufiev	2021	*Technological Forecasting and Social Change*	Russia	quantitative	secondary
Song, Wang, & Bergmann	2020	*International Journal of Information Management*	China	quantitative	secondary
van Ingen & Matzat	2018	*Information, Communication & Society*	Netherlands	quantitative	primary
Welser, Khan, Dickard	2019	*Information, Communication & Society*	USA	quantitative	primary
Yu, Lin & Liao	2017	*Computers in Human Behavior*	Taiwan	quantitative	primary
Zhang, Zhao & Qiao	2020	*Transport Policy*	China	quantitative	primary

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